
Supplementary information

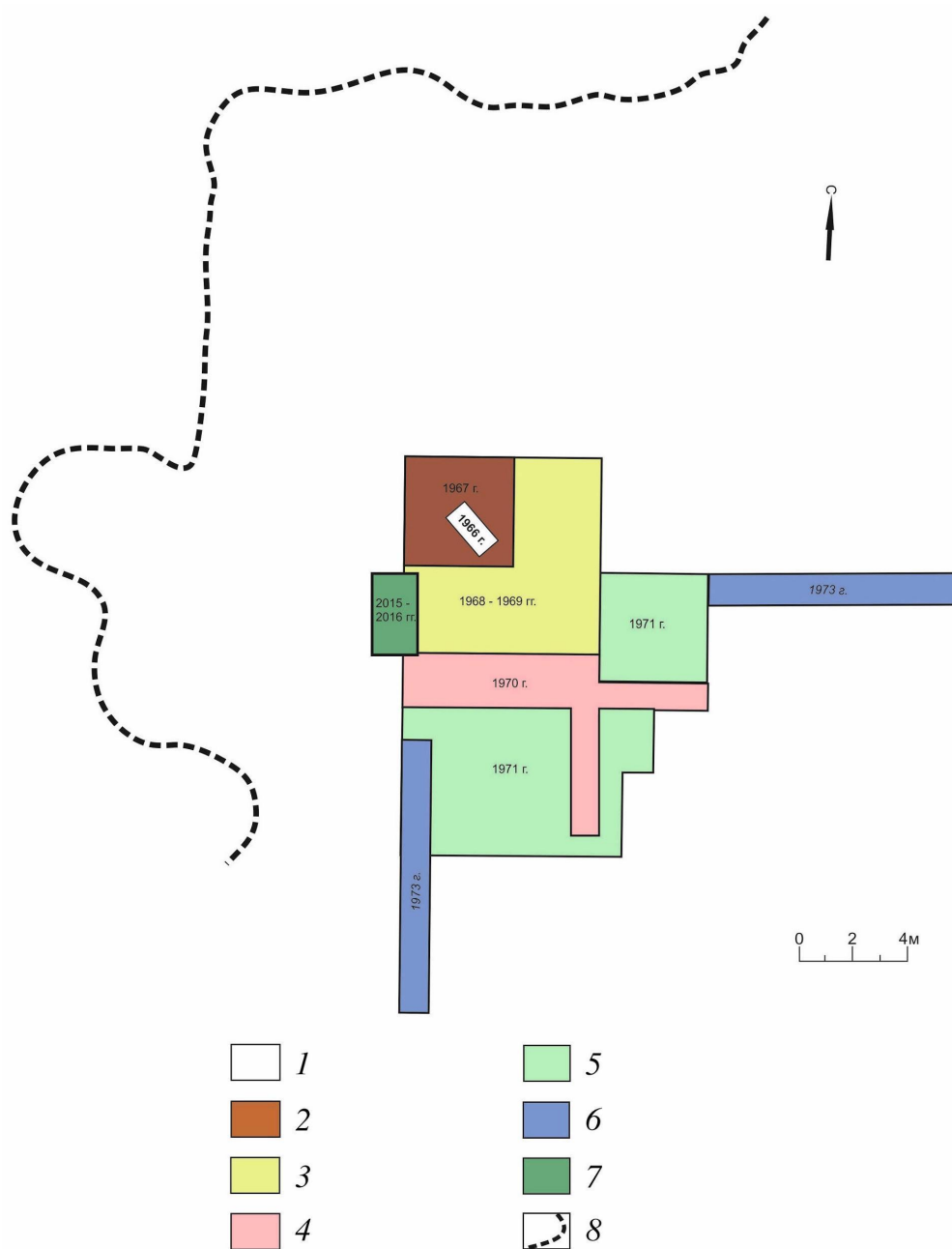
Evidence for early dispersal of domestic sheep into Central Asia

In the format provided by the
authors and unedited

**Evidence for Early Dispersal of Domestic Sheep into Central
Asia: genetic, archaeozoological, and biomolecular insights
from Obishir V**

Supplementary Information

Supplementary Appendix A. Excavation



Supplementary Figure 1. Plan of excavation areas (A) at Obishir V site. A: 1 – 1966 probe trench; 2 – 1967 excavation; 3 – 1968–1969 excavation; 4 – 1970 excavation; 5 – 1971 excavation; 6 – 1973 trench; 7 – 2015–2016 excavation; 8 – internal boundary of the rockshelter.

Geological characteristics of sediments and stratigraphy are presented in Supplementary Table 1. Nine stratigraphic units (layers or sublayers) were recognized, although the lithological variability is low. The entire sedimentary sequence was influenced by rockfall and scree accumulation at the base of the steep hill with outcrops of Paleozoic limestones and shales. Thus, the sediments are mostly represented by a high amount of angular and subangular clasts of different size. This facies is typical of slope deposits connected with cold and dry conditions (Pawelec 2011, Pawelec et al. 2015). The sequence may be subdivided into three lithostratigraphic series. Series I includes Layer 5. This lowermost stratum is an eolian loess-like deposit. Series II includes colluvial sediments of Layers 4, 3, 2.4, 2.3, 2.2, up to 2.1. These strata were deposited in milder climatic conditions and contain material re-deposited down the slope from the area of the rockshelter niche, toward SE. The primary deposition of sediments inside the rockshelter and their re-deposition were diachronic. Radiocarbon dates achieved for charcoals and bones (Supplementary Table 2) most probably represent the age of primary deposition, which lasted for more than two millennia. The deposition of this series finalized with the last phase of anthropogenic accumulation recorded in Sublayer 2.1 and pedogenesis. Series III includes layers 1 and 0, deposited as an effect of the anthropogenic and biogenic activity during the last centuries.

References cited

Pawelec H. 2011. Periglacial evolution of slopes – Rock control versus climate factors (the Cracow Upland, Poland). *Geomorphology* 132, 139-152

Pawelec H., Drewnik M., Żyła M. 2015. Palaeogeography and lithostratigraphy of slope deposits in the loess area (the Miechów Upland, Poland) based of facies and microfacies analyses. *Geomorphology* 232, 145-163

Supplementary Table 1. Description of sediments from Obishir V.

Layer	thickness (cm)	Macroscopic lithological description	Genetic interpretation	Climate interpretation
0	30	dark gray silt with plant roots and litter	recent topsoil	moderately humid
1	60	light gray to grayish brown (dry 10YR 7/1, moist 2.5Y 5/2) silt with mid- to coarse-grained limestone clasts (chaotic texture), with pottery shards, bones and charcoals; with fireplaces and anthropogenic pits; lower boundary clear, marked with change in color and texture	cultural anthropogenic pedosediment; bioturbation	dry
2.1	5	light gray to pale brown (dry 10YR 7/2, moist 10YR 6/3) thin-laminated compacted silt; lower boundary clear, inclined, sedimentary	sheet wash by rainwater/meltwater, followed by pedogenesis and trampling (living floor)	moderately humid, or catastrophic event with increased water supply
2.2	40	pinkish gray to yellowish brown (dry 7.5YR 7/2, moist 10YR 5/4) loose silt with sparse fine-grained limestone clasts; lower boundary sharp, discordant	original source sediment deposited as rock fall with participation of anthropogenic deposition; colluvial re-deposition followed by pedogenesis	moderately humid
possible hiatus				
2.3	60	light brownish-gray to dark brown (dry 10YR 6/2, moist 10YR 3/3) silty loam with very abundant fine- to mid-grained limestone clasts (planar texture) and fine pebbles (0.5-10 cm); with abundant lithic artefacts; lower boundary inclined, erosional, marked with mid- and coarse-grained limestone clasts	original source sediment deposited as rock fall with participation of anthropogenic deposition; rill erosion followed by colluvial re-deposition;	humid, or catastrophic event with increased water supply, then moderately humid
hiatus				
2.4	40	light gray to brown (dry 10YR 7/2, moist 10YR 4/3) silty loam with very abundant fine- to mid-grained limestone clasts (planar texture) and fine pebbles (0.5-1 cm); lower boundary inclined, discordant, marked with mid- and coarse-grained limestone clasts	colluvial re-deposition	moderately humid
possible hiatus				

3	90	very pale brown to yellowish-brown (dry 10YR 7/3, moist 10YR 5/4) silt with small amount of fine- to mid-grained limestone clasts (chaotic texture); lower boundary inclined, discordant, marked by accumulation of coarse-grained limestone clasts	Erosion followed by colluvial re-deposition	moderately humid
possible hiatus				
4	110	very pale brown to yellowish-brown (dry 10YR 7/3, moist 10YR 5/4) silt with small amount of fine- to mid-grained limestone clasts (chaotic texture); lower boundary unclear, gradual passing into layer 5	accumulation of loess and rock fall; followed by erosion and colluvial re-deposition	cold and dry, then moderately humid
possible hiatus				
5	70	pale brown to yellowish-brown (dry 10YR 6/3, moist 10YR 5/5) hard silt, massive texture, almost no limestone clasts; preserved only in depressions of the bedrock	accumulation of loess and rock fall	cold and dry

Supplementary Appendix B. Radiocarbon and thermoluminescence dating

Section B1. OxCal code used in the analysis

Outlier model – no charcoal dates

```
Plot()
{
  Outlier_Model("General",T(5),U(0,4),"t");
  Sequence("Obishir")
  {
    Boundary("Layer 5 Start");
    Phase("Layer 5")
    {
      Date("UG7089",N(2015-12200,1900))
      {
        Outlier("General", 0.05);
      };
      Date("UG7090",N(2015-12400,1900))
      {
        Outlier("General", 0.05);
      };
      Date("Layer 5");
      Sum("Obishir Layer 5");
    };
    Boundary("Layer 5 End");
    Boundary("Layer 4 Start");
    Phase("Layer 4")
    {
      R_Date("GrM-11871 bone", 8842, 18)
      {
        Outlier("General", 0.05);
      };
      Date("Layer 4");
      Sum("Obishir Layer 4");
    };
    Boundary("Layer 4 End");
    Boundary("Layer 2-3 Start");
    Phase("Layer 2-3")
    {
      R_Date("GrM-11929 bone", 7257, 19)
      {
        Outlier("General", 0.05);
      };
      R_Date("GV 2118 tooth", 7324, 105)
      {
        Outlier("General", 0.05);
      };
      R_Date("GV 2119 tooth", 7499, 95)
      {
        Outlier("General", 0.05);
      };
      R_Date("OxA-37834 bone", 7100, 40)
      {
        Outlier("General", 0.05);
      };
      R_Date("GRM-12293 bone", 8005, 30)
      {
        Outlier("General", 0.05);
      };
      R_Date("AA-114469 tooth", 6989, 45)
      {
        Outlier("General", 0.05);
      };
    };
  };
};
```

```

R_Date("AA-114606 tooth", 7012, 45)
{
  Outlier("General", 0.05);
};
R_Date("AA-114607 tooth", 4260, 36)
{
  Outlier("General", 0.05);
};
Date("Layer 2-3");
Sum("Obishir Layer 2-3");
};
Boundary("Layer 2-3 End");
Boundary("Layer 1 Start");
Phase("Layer 1")
{
  R_Date("AA-114608 tooth", 2796, 34)
  {
    Outlier("General", 0.05);
  };
  R_Date("OxA37833 bone", 1591, 26)
  {
    Outlier("General", 0.05);
  };
  Date("Layer 1");
  Sum("Obishir Layer 1");
};
Boundary("Layer 1 End");
};
};

```

Standard model – no charcoal dates

```

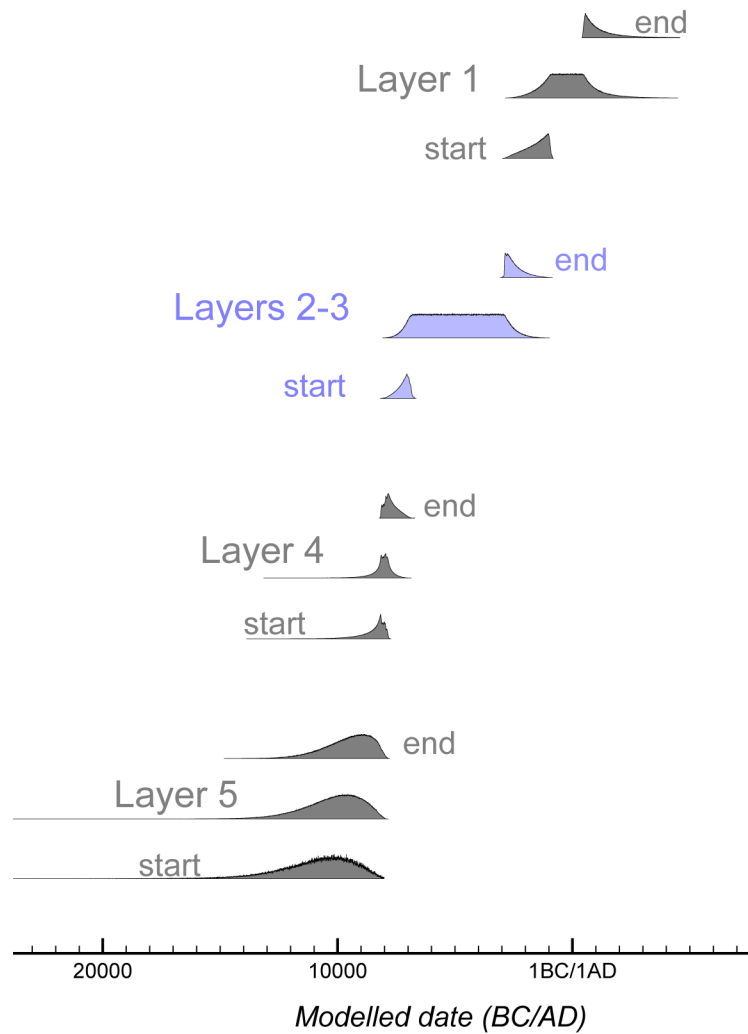
Plot()
{
  Sequence("Obishir")
  {
    Boundary("Layer 5 Start");
    Phase("Layer 5")
    {
      Date("UG7089", N(2015-12200, 1900))
      {
      };
      Date("UG7090", N(2015-12400, 1900))
      {
      };
      Date("Layer 5");
    };
    Boundary("Layer 5 End");
    Boundary("Layer 4 Start");
    Phase("Layer 4")
    {
      R_Date("GrM-11871 bone", 8842, 18)
      {
      };
      Date("Layer 4");
    };
    Boundary("Layer 4 End");
    Boundary("Layer 2-3 Start");
    Phase("Layer 2-3")
    {
      R_Date("GrM-11929 bone", 7257, 19)
      {
      };
      R_Date("GV 2118 tooth", 7324, 105)
    }
  }
}

```

```
{
};
R_Date("GV 2119 tooth", 7499, 95)
{
};
R_Date("OxA-37834 bone", 7100, 40)
{
};
R_Date("GRM-12293 bone", 8005, 30)
{
};
R_Date("AA-114469 tooth", 6989, 45)
{
};
R_Date("AA-114606 tooth", 7012, 45)
{
};
R_Date("AA-114607 tooth", 4260, 36)
{
};
Date("Layer 2-3");
};
Boundary("Layer 2-3 End");
Boundary("Layer 1 Start");
Phase("Layer 1")
{
R_Date("AA-114608 tooth", 2796, 34)
{
};
R_Date("OxA37833 bone", 1591, 26)
{
};
Date("Layer 1");
};
Boundary("Layer 1 End");
};
};
```

Section B2. Dates

Supplementary Figure 2. Modeled layer date ranges, along with start and end boundaries, using an ordered Bayesian radiocarbon model (uniform prior) for Obishir V, by layer, produced in OxCal using INTCAL20 calibration curve (Reimer et al. 2020).



Supplementary Table 2. Radiocarbon dates for Layers 1-4, Obishir V. Grey indicates date inconsistent with stratigraphy.

	Context	Material	Description	¹⁴ C Result	Error (+/-)	2 sigma calibrated range	Lab #
Cultural component 1	Obishir R-8 Layer 1 Depth 3 (19.32, #710)	Tooth	Equid lower R M2	1591	26	409-540 cal. CE	OxA-37833
	Obishir, R-10; Depth-278 Layer 1	Charcoal	Unidentified wood charcoal	1650	20	343-426 cal. CE	PLD-31751
Boundary	Obishir R-9, Layer 1/Layer 2 contact (21.4)	Tooth	Capra tooth	2796	34	1045-835 cal. BCE	AA-114608
Cultural component 2	Obishir R10/11 Layer 2 (20.6)	Tooth	Ovis tooth	7012	45	5991-5771 cal. BCE	AA-114606
	Obishir R-8/3 Layer 2 Depth 1 (4.10)	Bone	Mammal bone fragment	7100	40	6050-5899 cal. CE	OxA-37834
	Obishir T-11, Layer 2.1 Depth 1	Tooth	Sheep/goat tooth	7324	105	6406-6013 cal. BCE	GV-2118
	Obishir, R-10; Depth-325; Layer 2	Charcoal	Unidentified wood charcoal	7405	25	6367-6229 cal. BCE	PLD-31752
	Obishir, R-8; Layer 2.2, Depth 2	Tooth	Sheep/goat tooth (upper M3)	6989	45	5976-5764 ca. BCE	AA-114469
	Obishir R10, Layer 2 Depth 4 (20.4)	Tooth	Ovis tooth	4260*	36	3003-2701 cal. BCE	AA-114607
	Obishir R-9/2 Layer 2.3, Depth 15	Bone (tooth)	Ungulate tooth (Zooms ID: Ovis/Capra/cervid)	7499	95	6566-6105 cal. BCE	GV-2119
	Obishir R-10 Layer 2 Depth 7-8 (O1)	Bone	Mammal bone fragment	8005	30	7058-6821 cal. BCE	GrM-12293
	Obishir, R-11, Depth-352, Layer 2	Charcoal	Unidentified wood charcoal	9410	30	8770-8620 cal. BCE	PLD-31753
	Obishir R-9 Layer 3 Depth 2 (O3)	Bone	Mammal bone fragment - charred	7257	19	6210-6062 cal. BCE	GrM-11929
	Obishir R-8/3 Layer 4 Depth 4 (O2)	Bone	Mammal bone fragment (Zooms ID: Ovis)	8842	18	8186-7826 cal. BCE	GrM-11871
	Obishir R-10 Layer 4 Depth 19 (O4)	Bone	Mammal bone fragment	Failed	---		---

Supplementary Table 3. Thermoluminescence dates for Layer 5, Obishir-V.

Sample reference number	Depth	No lab. UG	²²⁶ Ra [Bq / kg]	²³² Th [Bq/kg]	⁴⁰ K [Bq / kg]	Dose rate d _r [Gy / ka]	Equivalent dose d _e Gy)	TL age [ka]
Obishir, I. 5.1	295 335	7089	25.2±2.4	22.0±2.0	451±45	2.68±0.26	32.7±3.4	12.2±1.9
Obishir, I.5.2		7090	25.9±2.6	25.7±2.6	489±49	2.75±0.28	34.1±3.4	12.4±1.9

Section B3. Human remains

Methods

The tooth sample was first cleaned overnight with 0.1M acetic acid. It was then thoroughly rinsed with demineralized water and dried. To perform the radiocarbon measurements, this sample was acid digested and measured in the form of gaseous CO₂ using the Groningen-MICADAS. Acid-digestion was performed on the Carbonate Handling System (CHS, Ionplus) and the liberated CO₂ was transferred in to the MICADAS using the Gas Interface System (GIS, Ionplus) (Wacker et al. 2013). Three portions of approximately 10 mg sample was weighed in to three screw-capped sampled vials (3.7 ml Exetainer® vials with Chlorobutyl septum) to perform triplicate measurements. The vials were flushed with He to remove the air contained in them. Then, 0.5 ml phosphoric acid (85%) was added into each of the vials, heated to 50 °C. To ensure complete liberation of CO₂ from the samples, the reaction was allowed to continue for approximately 12 hours at 50 °C. Following this, the produced CO₂ (~ 80 µgC) was flushed out with He and dried through a Sicapent® drying tube and transferred in to the GIS, where it was diluted with an appropriate amount of He before measurement was performed on the MICADAS. A 10 min measurement was performed on each sample. Data was analyzed using the standard BATS program provided with the MICADAS3.

Results

¹⁴C ages have been calibrated to calendar years with software program OxCal (Bronk Ramsey, 1995) version 4.4 using the IntCal20 calibration curve (Reimer et al. 2020).

Supplementary Table 4. Radiocarbon results from human tooth recovered from sieve at contact of Layers 1 and 2.

Name	Material (pretreatment)	Dated fraction	Result	F ¹⁴ C (± 1 sigma)	¹⁴ C Age (yr BP, ± s)	Calibrated age range (2 sigma)
Obishir V, human tooth	Tooth (HAc etching + H ₃ PO ₄ digestion)	Carbonate	Measurement 1	0.9008 ±0.0063	840 ±60	1185-1290 cal. CE
			Measurement 2	0.9112 ±0.0066	750 ±60	
			Measurement 3	0.9160 ±0.0071	710 ±65	
			GrM-20754	0.9087 ±0.0038	770 ±40	

References cited

Wacker, L.; Fülöp, R. H.; Hajdas, I.; Molnár, M.; Rethemeyer, J. 2013. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* 294, 214-217.

1
insufficient collagen
for identification

cervidae
cervidae
bovidae 1

7
deer
Gazella sp.
Saiga sp.
ovinae

17
Ovis sp.

5
Capra sp.

4
Bos sp.

1
Equus sp.

1
Canis sp.
Vulpes sp.

Supplementary Figure 3. Collagen peptide-based identifications for early Middle Ages Layer 1, Square R8 at Obishir V, grouped into cladogram by general phylogenetic relationships. Specimens with a grey circle are missing necessary peptide markers for more detailed identification, and silhouettes depict a range of possible taxa that specimens in that category may derive from.

Supplementary Appendix D. Equid DNA analysis

Ancient DNA extraction and analysis

Sample preparation and DNA extraction were performed in the ancient DNA facilities of Laboratoire AMIS CNRS UMR 5288. We successfully extracted ancient DNA from individual AMIS_1_01048 following the protocol described in Yang et al. (1998) and modified as in Gamba et al. (2016), and incubated extraction DNA with USER™ enzyme mix (NEB) for three hours at 37°C. Subsequently, we prepared Blunt-End genomic libraries following Meyer and Kircher (2010), with the modifications from Rohland et al. (2015) regarding dual internal indexing, that we amplified via PCR and quantified on TapeStation 4200 instrument (Agilent technologies). DNA libraries comprised two internal indexes (7bp each) and one external index (6bp). The purified and quantified library was sequenced on an Illumina MiniSeq platform. We generated 978,378 sequences mapping to the horse reference nuclear and mitochondrial genomes, which enabled us to unambiguously identify this individual as a male donkey using the Zonkey pipeline dedicated for species and sex molecular identification.

Supplementary Table 5. DNA protocols used in equid tooth analysis from Obishir, Layer 1.

Sample ID	Bone powder (mg)	Extraction method	USER treatment	Library method	DNA polymerase	PCR cycles	Internal index	External index
AMIS_1_01048	310	Yang (1998) Gamba et al (2016)	yes	Meyer & Kircher (2010) Triple Index	Accuprime	12	TCGTCTA	ATGGTT

Supplementary Table 6. DNA results for equid tooth from Obishir, Layer 1.

Sample ID	Sequencing platform	Retained reads	Endogenous fraction	Unique reads (nuclear)	Unique reads (mitochondrial)	Nuclear coverage	Mito-chondrial coverage	Species	Sex
AMIS_1_01048	MiniSeq	978378	13.95%	135229	96	0.004	0.35	Donkey	Male

References cited

Gamba, C., Hanghøj, K., Gaunitz, C., Alfarhan, A.H., Alquraishi, S.A., Al-Rasheid, K.A.S., Bradley, D.G., Orlando, L., 2016. Comparing the performance of three ancient DNA extraction methods for high-throughput sequencing. *Mol. Ecol. Resour.* 16, 459–469.

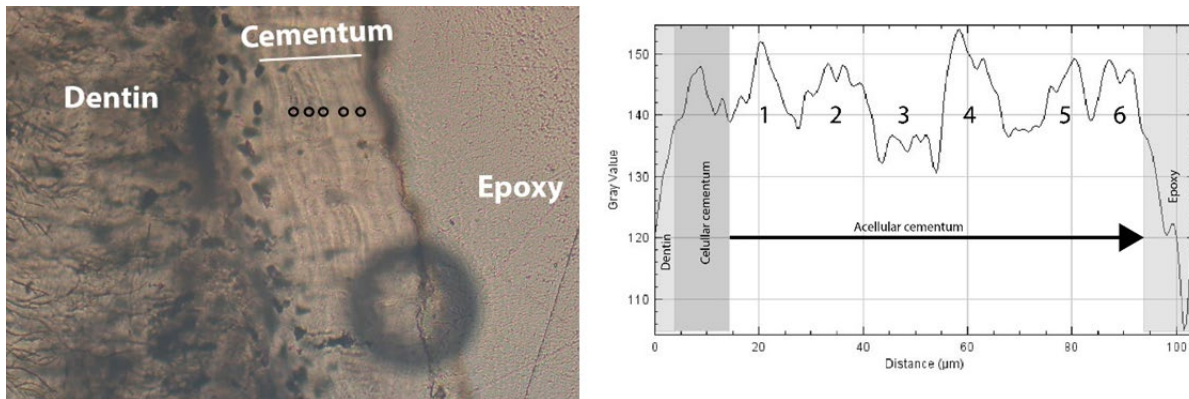
Meyer, M., Kircher, M., 2010. Illumina sequencing library preparation for highly multiplexed target capture and sequencing. *Cold Spring Harb. Protoc.* 2010, db.prot5448.

Rohland, N., Harney, E., Mallick, S., Nordenfelt, S., Reich, D. 2015. Partial uracil–DNA–glycosylase treatment for screening of ancient DNA. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 370, 20130624

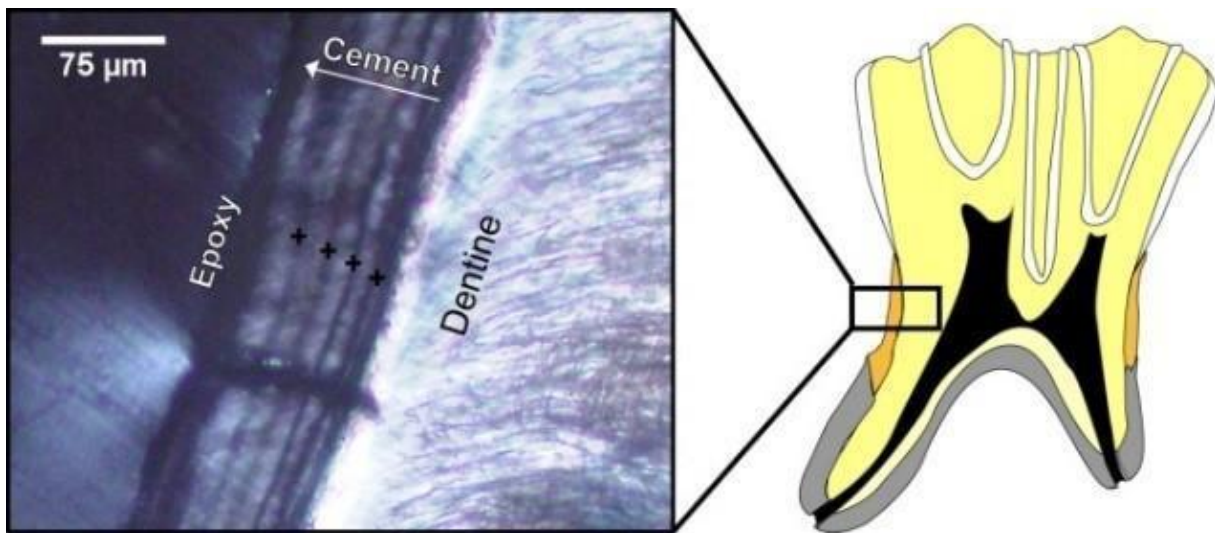
Schubert, M., Mashkour, M., Gaunitz, C., Fages, A., Seguin-Orlando, A., Sheikhi, S., Alfarhan, A.H., Alquraishi, S.A., Al-Rasheid, K.A.S., Chuang, R., Ermini, L., Gamba, C., Weinstock, J., Vedat, O., Orlando, L., 2017. Zonkey: A simple, accurate and sensitive pipeline to genetically identify equine F1-hybrids in archaeological assemblages. *J. Archaeol. Sci.* 78, 147–157.

Yang, D.Y., Eng, B., Wayne, J.S., Dudar, J.C., Saunders, S.R., 1998. Improved DNA extraction from ancient bones using silica-based spin columns. *American Journal of Physical Anthropology* 105, 539–543.

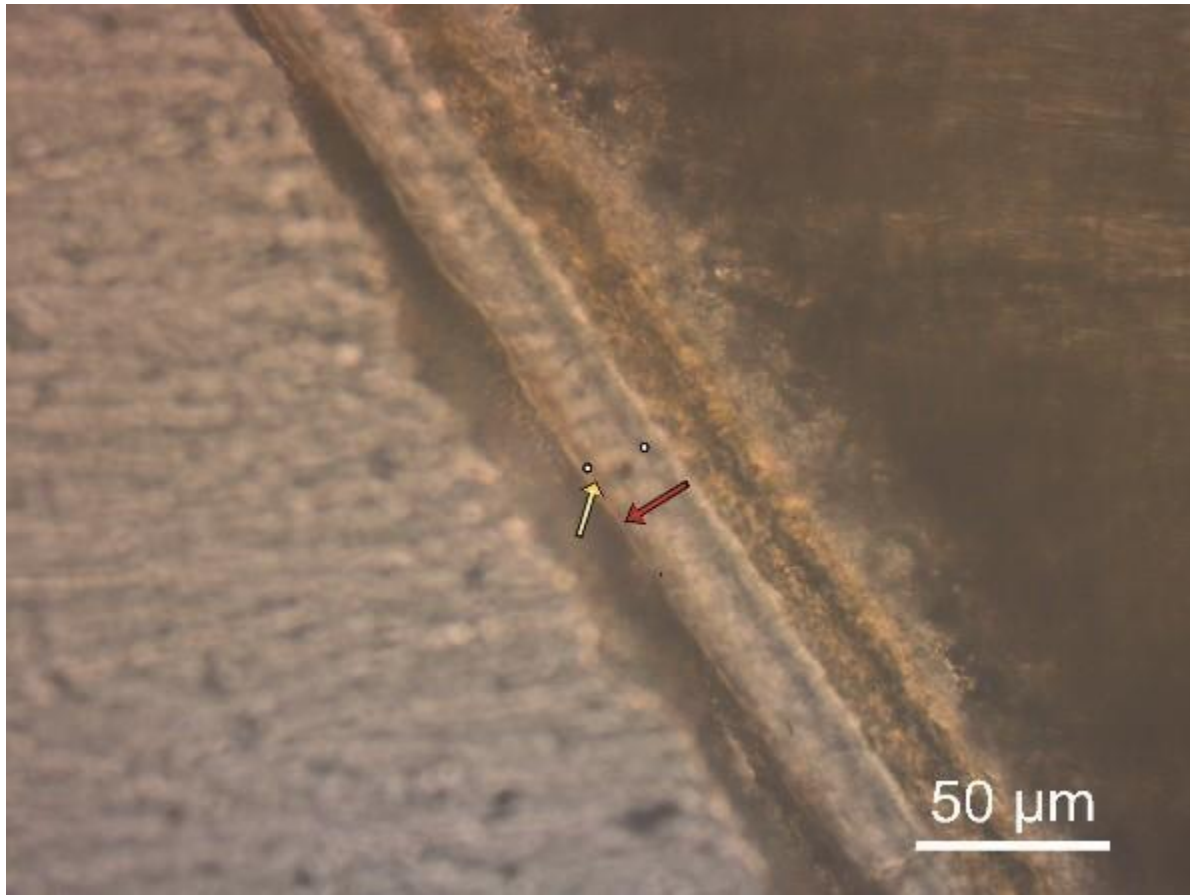
Supplementary Appendix E. Cementum Analysis



Supplementary Figure 4. Obishir 20.1 A, left: observation conducted under polarized light and high magnification (X500). The dots indicate annual banding B, right: Luminescent profile of the cementum, expressed in inverted greyscales. Growth zone appears as positive peaks. Numbers denote individual identified growth zones. Observations are conducted in mixed cementum. As shown on the profile, the last deposit has nearly the same dimensions as the five previous one, suggesting that the death occurred at the end of its formation.



Supplementary Figure 5: Schematic section of a ungulate tooth, with cementum shown in gray. The photograph shows bands of cementum observed under a polarizing microscope. The arrow indicates the direction of growth of the cementum, while annual banding is denoted with the '+' symbol.



Supplementary Figure 6. Obishir 20.6, observation conducted under polarized/analyzed light with insertion of lambda plate under high power magnification (X500). Dots show the location of annualized banding, and yellow shows the location of the outermost deposit, with red arrow indicating the direction of growth of the cementum.

Supplementary Appendix F. DNA analysis of *Ovis* and *Capra* specimens, Layer 2.

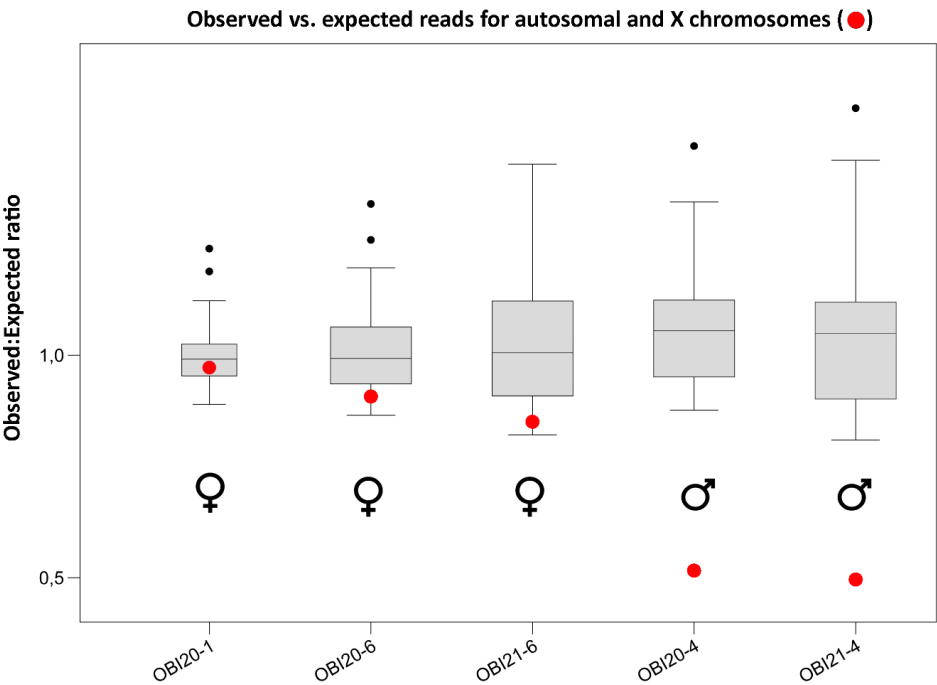
Unfortunately, the low endogenous content and low genome coverage of the late Bronze Age *Capra* specimen prevented nuanced taxonomic analyses using genome-wide data. However direct PCR amplification of the mitochondria cytochrome B (MT-CYB) gene and hypervariable region 1 (HVRI) allowed a species-level identification of *C. hircus* (Supplementary Table 7). Based on sequence alignment to 86 published wild and domesticated goats mitogenomes, OB21-04 PCR data are consistent with domestic goats and did not show any polymorphisms specific to modern wild goat taxa (*C. sibirica*, *C. falconeri* or *C. aegagrus*).

Supplementary Table 7. Whole mitogenome DNA results for specimens analyzed in this study.

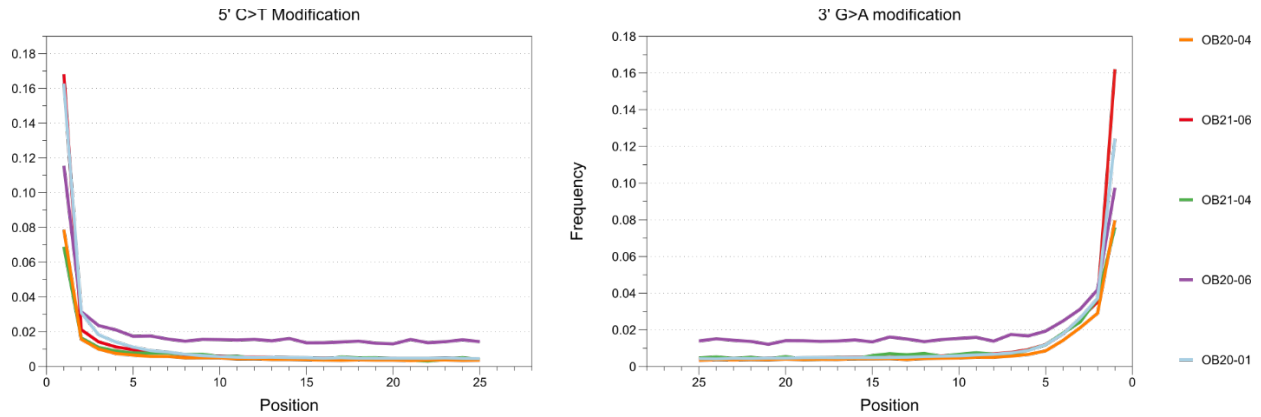
Specimen ID	Species	Reads sequenced	Endogenous DNA	# genome- wide SNPs	Sex	# mt reads	mt genome coverage	mt haplogroup
OB21-04	<i>Capra sp.</i>	27141767	0.27%	19,313	XY	37	0.3X	n.d.

n.d., not determined

**HVRI and MT-CYTB numbering for goat is relative to the goat reference genome KR059154.



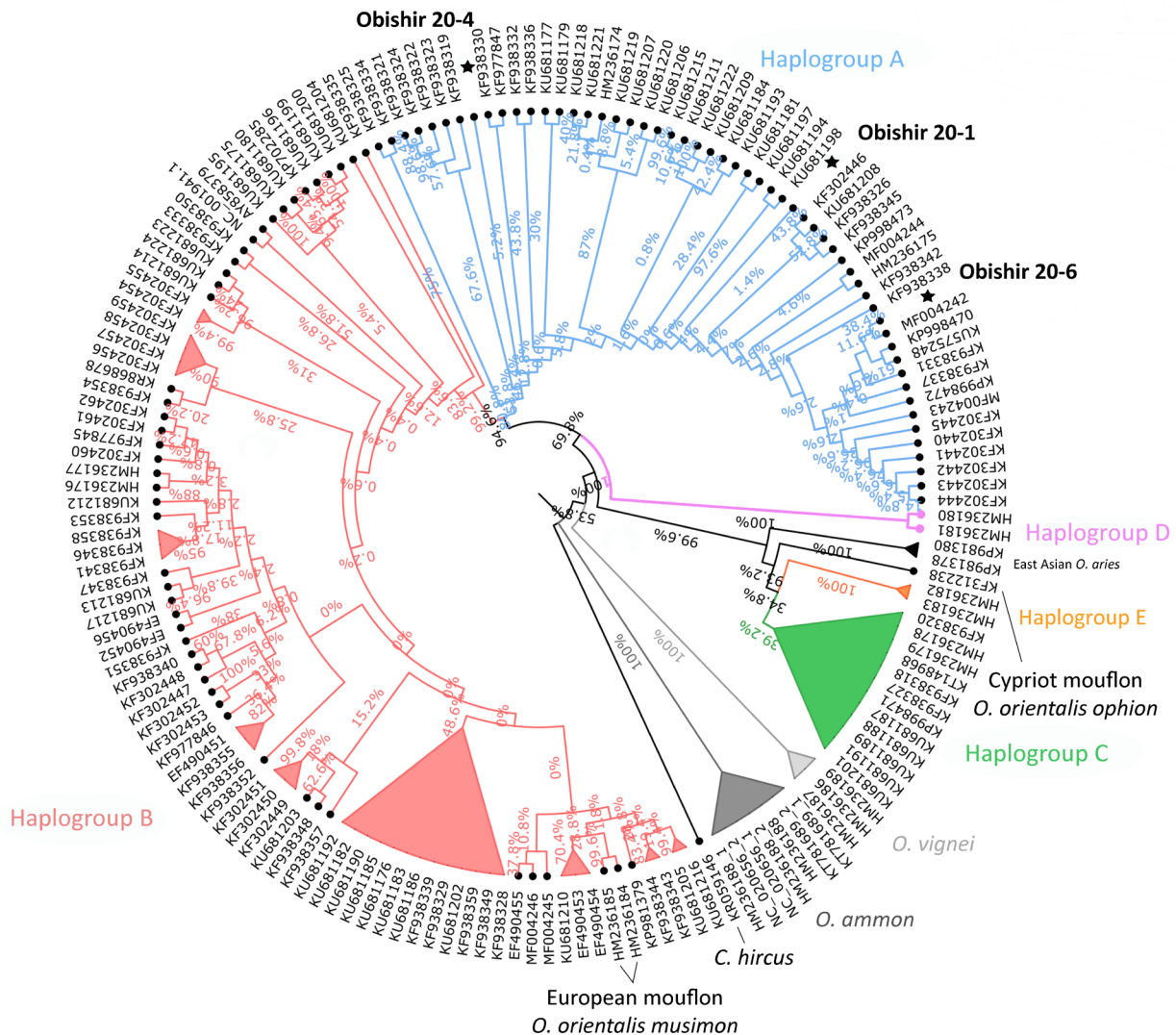
Supplementary Figure 7. Comparison of autosomal sequences (grey) to sex chromosome sequences (red) for analyzed samples. For female specimens, both numbers should cluster near a value of 1, while male specimens should exhibit a ratio of around 1 for autosomal sequences and a value of 0.5 for the X chromosome.



Supplementary Figure 8. Postmortem DNA damage patterns for Ovis and Capra specimens from Obishir V analyzed in this study after half-UDG treatment.

Maximum Parsimony analysis

Evolutionary history was inferred using the Maximum Parsimony method. The most parsimonious tree with length = 4457 is shown in the main text (Figure 6). The consistency index for this tree is 0.608000, the retention index is 0.901853, and the composite index is 0.663895 all sites and 0.54832) for parsimony-informative sites. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (500 replicates) are shown next to the branches [1]. The MP tree was obtained using the Subtree-Pruning-Regrafting (SPR) algorithm (2) with search level 1, in which the initial trees were obtained by the random addition of sequences (10 replicates). The tree is drawn to scale, with branch lengths calculated using the average pathway method [2] in the units of the number of changes over the whole sequence. This analysis involved 168 nucleotide sequences, with a total of 16858 positions in the final dataset. Evolutionary analyses were conducted in MEGA X [3,4].



Supplementary Figure 9. Phylogenetic tree produced using Maximum Likelihood showing the whole mitochondrial genome relationships of the Obishir V specimens (stars) to modern reference genomes of the domestic sheep *Ovis aries* (haplogroups A-E), the European mouflon sheep *O. musimon*, the Cypriot mouflon *O. orientalis ophion*, the wild *O. vignei*, and the wild argali sheep *O. ammon*, with *C. hircus* as an outgroup. Major haplogroups A, B, C, D and E were labeled following previous published studies.

Maximum Likelihood analysis

The evolutionary history was inferred by using the Maximum Likelihood method and Tamura-Nei model [5]. The tree with the highest log likelihood (-42641.06) is shown. The percentage of trees in which the associated taxa clustered together is shown next to the branches. Initial tree(s) for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the Maximum Composite Likelihood (MCL) approach, and then selecting the topology with superior log likelihood value. The tree (Supplementary Figure 9) is drawn to scale, with branch lengths measured in the number of substitutions per site. This analysis involved 164 nucleotide

sequences. There were a total of 16772 positions in the final dataset. Evolutionary analyses were conducted in MEGA X [3,4].

References cited

1. Felsenstein J. Confidence limits on phylogenies: An approach using the bootstrap. *Evolution* **39**:783-791 (1985).
2. Nei M. and Kumar S. *Molecular Evolution and Phylogenetics*. Oxford University Press, New York (2000).
3. S. Kumar, G. Stecher, M. Li, C. Knyaz, K. Tamura, MEGA X: Molecular Evolutionary Genetics Analysis across Computing Platforms. *Mol. Biol. Evol.* 35, 1547–1549 (2018).
4. G. Stecher, K. Tamura, S. Kumar, Molecular Evolutionary Genetics Analysis (MEGA) for macOS. *Mol. Biol. Evol.* 37, 1237–1239 (2020).
5. K. Tamura, M. Nei. Estimation of the number of nucleotide substitutions in the control region of mitochondrial DNA in humans and chimpanzees. *Mol. Biol. Evol.* 10, 512-526 (1993)

Supplementary Table 8. Primers used for PCR amplification. Positions are based on *C. hircus* mitogenome.

Name	Sequence	Region targeted	Size	Position
CAP-For	GATCTTCCYCATGCATATAAGCA	HV region	170bp	15938-16107
CAP-Rev	CGGGTTGCTGGTTTCAC			
CapF1	CTCTGTAACCTCACATTTGTCT	CytB	150bp	14340-14490
CapR1	GTTTCATGTTTCTAGAAAGGT			

Supplementary Table 9. Alignment of the CytB region for Ovis and Capra species

Consensus	1	10	20	30	40	50	60	70	80	90	98
1. KR059226 extraction (Capra aegagrus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
2. KR059210 extraction (Capra aegagrus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
3. KR059219 extraction (Capra aegagrus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
4. KR059221 extraction (Capra aegagrus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
5. KR059222 (Capra aegagrus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
6. FJ207525 extraction (Capra falconeri)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
7. NC_020622 extraction (Capra falconeri)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
8. KR059160 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
9. KR059162 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
10. KR059165 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
11. KR059164 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
12. KR059163 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
13. KR059168 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
14. KR059175 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
15. KR059223 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
16. KR059224 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
17. KR059225 extraction (Capra hircus)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
18. FJ207529 extraction (Capra sibirica)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
19. FJ207529 extraction (Capra sibirica)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
20. FJ207529 extraction (Capra sibirica)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
21. NC_020626 extraction (Capra sibirica)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
22. OB21-04-CAPR1B1_5.ab1	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
23. OB20-06-CAPR1B1_4.ab1	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
24. OB20-07-CAPR1B1_1.ab1	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
25. OB21-06-CAPR1B1_6.ab1	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
26. OB20-01-CAPR1B1_3.ab1	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
27. OB20-04-CAPR1B1_2.ab1	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
28. NC_001941.1 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
29. KU681178 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
30. KF938317 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
31. KU681210 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
32. HM236182 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
33. HM236178 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
34. KU681212 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
35. KU681215 (Ovis aries)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
36. NC_020656 (Ovis ammon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
37. NC_020656 (Ovis ammon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
38. HM236188 (Ovis ammon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
39. HM236188 (Ovis ammon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
40. AJ867274 (Ovis ammon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
41. AJ867274 (Ovis ammon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
42. HM236185 (Ovis aries musimon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
43. HM236184 (Ovis aries musimon)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT
44. KF312238 (Ovis orientalis ophion)	CTCTGTA	AACTCAC	-ATTTGTC	GAGATG	TAAAT	TATGGCT	GAATTAT	CCGATAC	ATACACG	CAACGG	GAGC-ATCAATAT

